

INVITED TOPICAL REVIEW

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Poststroke Depression and Anxiety: A Comprehensive Topical Review

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ABSTRACT: Mental health issues after stroke are common and have a significant negative impact on functional outcomes, quality of life, and longevity. Despite their prevalence and impact, these conditions remain underrecognized and undertreated. This topical review synthesizes current evidence on the epidemiology, mechanisms, assessments, and treatments related to poststroke depression and anxiety, the most common neuropsychological sequelae after stroke, and identifies gaps and future directions.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: anxiety disorders ■ cognition ■ depression ■ medication adherence ■ stroke



A dynamic, multifaceted set of biological, psychological, and social factors affects an individual's recovery trajectory and adjustment following stroke. Anxiety and depression are some of the most common complications, occurring in approximately one-third of all patients with stroke.^{1–3} Mental health needs change over time from acute to chronic to terminal phases and range from normal responses to a life-altering event to pathological conditions. Two separate but often co-occurring conditions, poststroke depression (PSD) and poststroke anxiety (PSA), have diagnostic criteria (Table 1) similar to nonstroke-related depressive and anxiety disorders; however, PSD and PSA are characterized by distinct mechanisms, symptom severity and patterns, and treatment responsiveness. For example, PSD manifests with less prevalent and less severe anhedonia compared with depression in the general population.⁵

The impact of PSD and PSA is substantial, including well-documented detrimental effects on functional independence, cognitive recovery, interpersonal relationships, community reintegration, participation in therapies, medication adherence, and quality of life. PSD is also associated with increased risk for stroke recurrence and mortality.^{6–8}

Since publication of the 2016 American Heart Association/American Stroke Association Scientific Statement for Healthcare Professionals on PSD,² mental health conditions poststroke, and in particular PSD, have received more attention. Yet, despite their prevalence and significant impact, mental health conditions remain underrecognized and undertreated. The purpose of this topical review is to synthesize current evidence on the epidemiology, mechanisms, assessment, treatment, and gaps related to PSD and PSA.

EPIDEMIOLOGY AND RISK FACTORS

Prevalence of Depression and Anxiety Poststroke

PSD is one of the most common neuropsychiatric sequelae and the most widely studied affective disorder following stroke, while PSA is also prevalent but less researched (see Table 2 for prevalence estimates). PSD and PSA frequently co-occur (odds ratio, 4.66),¹¹ further supporting the importance of assessing and treating both conditions. Importantly, stroke survivors have higher risks of suicidal ideation (12.2% versus 4.3%), suicide

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Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/STROKEAHA.125.051664>.
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Stroke is available at www.ahajournals.org/journal/str

Table 1. Diagnostic Criteria⁴ (Symptoms Lasting 2 Weeks or Longer)

PSD: depressive disorder due to another medical condition	PSA: anxiety disorder due to another medical condition
History of stroke	History of stroke
Depressed mood and irritability	Prominent anxiety
Anhedonia (lack of interest/pleasure in previously enjoyable activities)	Excessive fear, worry, and concern about health issues
Changes in appetite	Intense dread and uneasiness
Changes in sleep	Panic attacks
Psychomotor agitation or retardation	
Difficulties concentrating	
Lack of energy and fatigue	
Feeling hopeless and worthless	
Thoughts of suicide or death	
ICD-11: mood disorder due to a known physiological condition with depressive features. F06.31	ICD-11: anxiety disorder due to a known physiological condition is F06.4.

ICD-11 indicates *International Classification of Diseases, Eleventh Revision*; and PSD, poststroke depression.

attempts, and suicide deaths (73% greater) compared with the general population,^{12,13} highlighting stroke as a significant risk factor. Unfortunately, few patients are ever screened for suicidal ideation.

Reported prevalence varies by assessment timing and diagnostic approach, with structured interviews consistently yielding lower estimates than rating scales. Study quality should also be considered, as many studies exclude patients with severe impairments, likely underestimating prevalence in high-risk groups, including those with aphasia or profound disability. Thus, while prevalence statistics are useful for quantifying population-level burden, they often miss the nuanced, individualized experiences of poststroke mood and anxiety disturbances.

Risk Factors

The development of PSD and PSA reflects converging demographic, socioeconomic, and clinical influences that contribute to risk.

Demographic Factors

Research indicates that younger stroke survivors (<50 years, though definitions vary) are at greater risk for PSD and PSA.^{11,14} However, older adults may be underdiagnosed due to overlapping symptoms with cognitive decline or other comorbidities.¹⁵ Similar to nonstroke populations, studies also consistently show higher instances of PSD and PSA in women than in men.^{11,16} Though the literature on PSD by race/ethnicity remains sparse, existing research shows that Hispanic and Black stroke survivors have a higher prevalence of PSD than non-Hispanic White stroke survivors.^{17,18} Furthermore, Black people with PSD are at significantly greater risk of 3-year stroke recurrence than White people with PSD.¹⁹

Table 2. Prevalence of PSD and PSA by Assessment Method, Timing, and Age^{1,9,10}

Prevalence category	PSD	PSA
Overall	27%	18.7%–24.2%
Using structured clinical interview	24%	18.7%
Using rating scales	29%	24.2%
During early poststroke period	Within 1 mo: 30%	Within 1 mo: 15.5% (interview) and 25.5% (rating)
During later poststroke period	>1 y: 29%	>2 y: 20.4% (interview) and 26.0% (rating)
Among young stroke survivors	31%	39%

PSA indicates poststroke anxiety; and PSD, poststroke depression.

Barriers in access to poststroke and mental health care, such as lack or type of insurance, language, stigma, and other sociodemographic factors, likely contribute to these disparities.

Socioeconomic Factors

Financial difficulties, unemployment, low educational attainment, and reduced access to health care, including outpatient rehabilitation and other specialty services, can contribute to poor mental health poststroke.^{11,20,21} Socioeconomic adversity likely amplifies risk through reduced access to care and chronic stress exposures; however, robust prospective cohort studies are needed to better quantify independent effects. Conversely, the presence of an engaged and stable caregiver network is a protective factor.¹¹

Clinical Factors

Stroke severity is a pivotal clinical determinant of psychological outcomes. More severe strokes typically result in greater functional impairment, strongly correlating with higher rates of both PSD and PSA.^{11,22} Impaired communication and cognition contribute significantly to PSD and PSA, likely by fostering feelings of isolation and frustration.^{20,23,24}

Prestroke depression and anxiety nearly triple PSD risk, and a family history of mental illness nearly doubles the risk.^{11,25} Preexisting medical comorbidities may increase PSD risk to some extent,^{11,26} whereas fatigue and sleep disturbance are more strongly associated with PSD and PSA risk.¹¹

PATHOPHYSIOLOGY AND MECHANISMS

The cause of mental health issues poststroke is likely multifactorial, consisting of biological and psychosocial factors that interact dynamically and change across time.

Neurobiological Factors

Primary pathophysiological mechanisms implicated in poststroke neuropsychiatric sequelae include disrupted

neurocircuitry, monoamine dysregulation, heightened neuroinflammatory responses accompanied by hypothalamic-pituitary-adrenal (HPA) axis dysfunction, compromised neurotrophic signaling, and genetic polymorphisms.³

Lesion Location and Neurocircuitry

Key regions involved in depression include the basal ganglia and limbic cortex (amygdala and hippocampus), the cingulate cortex, dorsal striatum, and the frontal lobe (dorsolateral prefrontal and medial prefrontal cortex). Patients with large infarct volumes, especially in the frontal lobe and basal ganglia, are more likely to develop PSD.³ Although it is commonly believed that PSD is associated with left hemisphere strokes, more recent reviews of the literature conclude that the relationship between lesion laterality and PSD is much more mixed.^{2,3,27–29} Advances in technology have allowed more investigation in this area but continue to yield inconsistent results and are based on small sample sizes. Similarly, evidence has not been able to draw definitive connections between lesion location or laterality and PSA.³⁰

Monoamine Dysregulation

Monoaminergic neurotransmitters are critical modulators of central nervous system function, influencing mood regulation, motivational states, arousal, and cognitive processing. Reduced levels of dopamine, norepinephrine, and serotonin contribute to PSD due to their central role in affective and executive functioning.^{3,31}

Neuroinflammation and HPA Dysregulation

Stroke elicits an acute inflammatory response involving increased levels of proinflammatory cytokines, resulting in glucocorticoid release. Neuroinflammation serves both protective and detrimental roles in stroke recovery, helping to clear necrotic tissue while also potentially exacerbating tissue damage.³² PSD is influenced significantly by these inflammatory processes, with microglial activation and cytokine release being key factors.³² Elevated proinflammatory cytokines, such as IL-1 β , IL-6, and TNF- α (tumor necrosis factor- α), are associated with a higher risk and severity of PSD.^{33,34} Conversely, although IL-10, an anti-inflammatory cytokine, had previously been thought to act as a protective factor against PSD,³⁵ a recent meta-analysis showed no significant effects.³³ If the adaptive acute inflammation becomes chronic, HPA axis dysregulation results, which can also contribute to the development of PSD.^{3,36} Stroke-related neuronal injury activates the HPA axis, resulting in elevated levels of corticotropin-releasing hormone and adrenocorticotrophic hormone. This activation is further intensified by proinflammatory cytokines, disrupting the HPA axis's regulatory negative feedback loop, which leads to sustained glucocorticoid elevation and reduced serotonin and norepinephrine production. Altogether, abnormal poststroke inflammatory response and higher glucocorticoid levels

have been associated with PSD, potentially through reduction in the transcription of neurotrophic factors and subsequent decreases in neurogenesis and neuroplasticity (particularly in the hippocampus and frontal cortex).^{3,27}

Neurotrophic Hypothesis and Neuroplasticity

BDNF (brain-derived neurotrophic factor) is a neurotrophin that plays a key role in neuronal repair after stroke and in mood disorders by promoting neuronal plasticity and influencing changes in the neural-immune axis. Notably, low-serum BDNF levels are linked to an increased risk and greater severity of PSD, but not PSA, potentially due to its neuroprotective and neurorestorative effects and influence on neurotransmitter systems.³⁷ Boosting BDNF levels, for example, through exercise or certain treatments, shows promise in alleviating PSD symptoms.³⁸

Genetics and Epigenetics

There is a growing body of evidence that suggests that genetic and epigenetic factors can influence the risk of developing PSD.³⁹ Specific polymorphisms in genes related to serotonin transport and neurotrophic factors are associated with increased risk of PSD.³ Understanding these genetic factors may help in developing personalized treatment strategies.

Psychosocial Factors

Alongside and interacting with neurobiological factors, psychosocial mechanisms play a significant role in the development and persistence of depression and anxiety poststroke. Stroke sequelae, including reduced mobility and impaired cognition and communication, can result in a loss of independence and profound changes in daily lives, relationships, and established social roles. For example, a sudden shift from being the primary provider to losing employment or relying on caregiver support for routine tasks can lead to feelings of being a burden and contribute to emotional distress. Having difficulties with mobility and communication can make it more challenging to engage socially, and the resulting perceived loneliness increases depression risk.⁴⁰ Cognitive distortions and maladaptive coping strategies, as well as preexisting psychological vulnerabilities, may interact with cognitive changes, further exacerbating risk. Stroke-related stigma also increases the risk of PSD, while stigma about mental health may prevent survivors from seeking treatment.⁴¹ Altogether, the challenge of adjusting to a new identity poststroke, combined with limited access to resources, creates a psychosocial environment conducive to mental health issues.

It is important to recognize that these factors do not occur in isolation, and that some of the associations between factors and mental health are bidirectional and interact. For example, greater functional impairment is a

risk factor for PSD; however, PSD can also contribute to greater disability due to poor medication and therapy adherence; poor maintenance of healthy lifestyle factors, such as nutrition, exercise, and sleep habits; and maladaptive coping (eg, use of alcohol), setting up a self-reinforcing cycle.

ASSESSMENT

Assessment of PSA and PSD involves a combination of screening questionnaires, clinical diagnostic interviews, and corroborative information from available family members and staff when appropriate.⁴² Consideration of setting (inpatient, subacute, and outpatient) and phase of recovery is necessary for ensuring assessment accuracy and reliability. Aphasia and cognitive impairment can also make assessment significantly more challenging as changes in communication ability,⁴³ memory, and awareness may limit the validity and reliability of information provided by the patient.⁴⁴ Importantly, PSA and PSD do not necessarily only present shortly after stroke,^{9,10} making routine assessment a priority.⁴⁵

Maintaining an appreciation of biological and psychosocial influences is necessary for differentiating between mental health and stroke-related symptoms.^{46,47} It is not uncommon for people with stroke to present with symptoms that seem like anxiety or depression² and for PSD and PSA to coexist with other neuropsychiatric syndromes after stroke, including fatigue, apathy, and emotional lability (pseudobulbar affect). Careful attention to core emotional symptoms, motivational state, and affective congruence can help differentiate these entities.

Poststroke fatigue is a common and often independent syndrome. Although no universally accepted definition exists, poststroke fatigue is typically described as an overwhelming sense of physical or mental exhaustion unrelated to exertion and not relieved by rest, or as a perceived lack of energy that interferes with daily activities. It may manifest physically (limiting endurance and activity) or cognitively, impairing concentration, multitasking, and stress tolerance. The presence of nonsomatic symptoms of emotional distress, such as persistently depressed mood and anhedonia, can help differentiate presentations of depression from poststroke fatigue.⁴⁸ Similarly, depressed mood, feelings of guilt, and worthlessness are associated with PSD, while poststroke apathy is characterized by a blunted emotional experience, decreased initiative, and goal-directed behaviors without emotional distress.⁴⁹ Pseudobulbar affect, a condition characterized by reduced voluntary control of affective expression such as laughing or crying is sudden, unpredictable, and incongruent with an individual's subjective emotional experience, distinguishing it from anxiety and depressive disorders with congruent affect and mood.⁵⁰ Other differential diagnoses include thyroid and other neuroendocrine disorders, sleep apnea, cardiovascular

issues, chronic pain, and cognitive impairment.⁵¹ Pharmacological side effects and other medical conditions may also overlap with common mental health symptoms, making accurate assessment challenging. For providers working in inpatient environments, delirium is a common and often underappreciated condition that impacts approximately 1 in 4 patients with stroke and can be easily mistaken for anxiety or depression, given both hyperactive and hypoactive presentations.⁵² Regarding mental health differential diagnoses, posttraumatic stress disorder (PTSD) is a condition in response to a life-threatening event (in this case, a stroke) that warrants special attention. Though estimates vary, poststroke PTSD occurs in up to 37% of people with stroke and is often comorbid with PSA or PSD.⁵³ With the publication of the Diagnostic and Statistical Manual of Mental Disorders-Fifth Edition,⁴ PTSD is no longer considered an anxiety disorder and has been moved to a new trauma- and stressor-related disorders category. While many symptoms of PTSD overlap with PSA and PSD, PTSD can be distinguished by the presence of intrusive symptoms, including recurrent intrusive memories, distressing dreams, dissociation reactions (eg, flashbacks to emergency department experiences, or perceived imminence of death), and prolonged psychological and physiological distress associated with trauma reminders. In addition, people with poststroke PTSD demonstrate persistent avoidance and dissociative symptoms associated with the trauma, which can lead to avoidance of medical settings or medications, whereas other anxiety conditions are not directly trauma-related. Awareness and consideration of symptom onset, duration, and variation, comorbid health conditions, potential precipitants, medical and psychiatric history, presence of other mental health symptoms or substance abuse, and response to interventions may help distinguish PSA and PSD from common stroke-related sequelae or other mental health conditions.^{51,54}

Early screening (within 3 months poststroke) for PSD has been associated with enhanced functional recovery, improved quality of life, and reduced mortality.⁵⁵ Repeated screening can help providers evaluate emerging mental health issues, assess the effectiveness of current interventions, and normalize the process of discussing mental health poststroke. When choosing a screening measure, administration time, staff availability, communication limitations, validation among people with stroke, and the degree of cognitive impairment should be considered.

Anxiety and Depression Screening

Screening tools can be divided into 3 categories: self-report questionnaires, clinician or caregiver/proxy-rated scales, and questionnaires designed for use in specific populations (eg, older adults and people with aphasia).⁵⁶ The 2-item Patient Health Questionnaire is the

most recommended brief screening tool for PSD due to its psychometric soundness and ease of implementation in a clinical setting.⁵⁷ There are notably fewer anxiety screening measures validated for stroke, which is consistent with less research and attention focused on PSA⁵⁸; however, the 2-item Generalized Anxiety Disorder offers similar advantages as the Patient Health Questionnaire-2. In clinical settings, having a measure for people with aphasia should also be considered. An exhaustive list of potential instruments is beyond the scope of this article; however, some of the most commonly referenced and validated measures for clinical and research use are briefly reviewed along with important considerations (Table S1).

Self-report measures, though most commonly used, may not be appropriate for people with aphasia or cognitive impairment, and some patients with physical limitations may need assistance to complete them. Clinician-rated measures, such as the Hamilton Depression Rating Scale,⁵⁹ may be limited by a lengthier administration time and may be more appropriate for a clinical interview than as a standard screening measure for all patients with stroke. To accommodate communication deficits, several aphasia-specific measures have clinicians or caregivers rate observable anxiety or depressive symptoms. The only available self-report measure for people with aphasia is the Visual Analog Mood Scale, a 5-minute clinician-administered screen assessing 8 mood states (afraid, confused, sad, energetic, tired, happy, and tense).⁶⁰ Last, measures are available for screening older adults, utilizing a more basic agree/disagree format, which may also be beneficial for people with nonfluent aphasia.

While there are several anxiety and depression screening options available, each of these measures comes with its own limitations. For example, many screening measures do not ask about suicide, a significant concern after stroke.^{12,13} In addition, many of these measures have only a few validation studies in stroke,⁵⁶ and the degree of cognitive or communication impairment present in research samples is often unclear, limiting generalizability.² Recent research has favored the use of the 9-item Patient Health Questionnaire and the Hamilton Depression Rating Scale; however, practical factors such as setting, phase of stroke recovery, and inclusion of people who may require aphasia-appropriate measures need to be considered. Providers need to be aware of how stroke-related cognitive, physical, and communication changes may impact the accuracy of screening measures and how bias may influence observer-rated measures. In general, screening is only one part of PSA and PSD assessment. Given these limitations, there are efforts to create improved screening measures for PSA and PSD although they are still in the early phases of development and validation.^{61,62} Finally, newer person-centered standardized short-form measures and computer adaptive

tests based on item banks from measurement systems such as PROMIS⁶³ and NeuroQOL⁶⁴ may be considered.

TREATMENT AND MANAGEMENT APPROACHES

Pharmacological Treatment

Poststroke Depression

Evidence on optimal pharmacological treatment duration and the choice of specific agents remains limited.⁶⁵ In addition, heterogeneity in diagnostic criteria, outcome measures, and stroke subtypes complicates direct comparisons between studies.³ Nevertheless, current guidelines support the use of antidepressants in patients with PSD, with the choice of agent based on clinical factors, side-effect profile, and pharmacological properties.² Ongoing monitoring should address both subtle and overt depressive symptoms while remaining alert to side effects and interactions, given the high prevalence of polypharmacy in stroke survivors. Treatment should continue for at least 6 months, with tapering planned carefully to avoid withdrawal. Dose adjustments are best guided by clinical response and tolerability, starting low and titrating cautiously.

Selective serotonin reuptake inhibitors (SSRIs) are considered the first-line agents for PSD because of their minimal anticholinergic activity and low risk of clinically significant drug interactions.⁶⁶ Sertraline, citalopram, and escitalopram are preferred, whereas paroxetine is less favored due to its stronger anticholinergic effects and higher interaction potential. Early studies (FLAME) suggested that fluoxetine might enhance motor recovery,⁶⁷ but recent large randomized controlled trials (RCTs; FOCUS, AFFINITY, and EFFECTS) demonstrated no functional benefit and an increased bone fracture risk, so routine prophylactic use for PSD prevention is not recommended.⁶⁸

For patients who do not tolerate or respond to SSRIs, serotonin-norepinephrine reuptake inhibitors such as venlafaxine and duloxetine represent reasonable alternatives. Mirtazapine, a heterocyclic antidepressant, may also be considered; however, studies have raised concerns regarding increased risks of seizures and cardiovascular events. Tricyclic antidepressants, including nortriptyline and amitriptyline, can be effective but are rarely used as first-line agents due to their anticholinergic and adverse cardiovascular profile.⁶⁹ Monoamine oxidase inhibitors (phenelzine, tranylcypromine, and isocarboxazid) are not recommended due to safety concerns, including tyramine-induced hypertensive crises, extensive drug interactions, and poorer tolerability compared with SSRIs and serotonin-norepinephrine reuptake inhibitors⁷⁰ (Table S2).

Other agents have a more limited or investigational role. Bupropion, a norepinephrine-dopamine reuptake

inhibitor, offers advantages with respect to weight and sexual side effects, but current evidence in PSD is insufficient to recommend routine use.⁷¹ Psychostimulants (methylphenidate) have been explored in inpatient settings to achieve rapid symptom relief or to augment SSRIs, particularly in patients with fatigue or cognitive impairments.² However, concerns about cardiovascular risk and reversible cerebral vasoconstriction underscore the need for larger RCTs before their role can be defined. Ketamine and esketamine are FDA-approved for treatment-resistant major depressive disorder; however, no clinical trial evidence is currently available to support their use in PSD though several registered trials are ongoing.⁷² Preclinical studies in rodent models of PSD have shown encouraging effects, including anti-inflammatory activity, enhanced synaptic plasticity, modulation of BDNF signaling, and upregulation of neurotrophic pathways in key brain regions.⁷³ These findings highlight their potential as future therapeutic options for PSD.

Poststroke Anxiety

SSRIs and serotonin-norepinephrine reuptake inhibitors are generally considered first-line pharmacological treatments for PSA due to their demonstrated efficacy, tolerability, and favorable safety profile. Benzodiazepines (alprazolam, clonazepam, diazepam, and lorazepam), in contrast, are generally discouraged because of their potential for dependence and deleterious effects, including cognitive impairment and psychomotor disturbances, increasing risk of falls and fall-related injuries, and higher short-term mortality.^{74,75} Age-related changes in drug metabolism, polypharmacy, and multiple comorbidities further increase susceptibility to adverse effects in this population. For older stroke survivors, these risks are compounded because sudden discontinuation can precipitate withdrawal symptoms, including rebound anxiety, insomnia, depression, seizures, or delirium, reinforcing the need for extreme caution when considering benzodiazepines as anxiolytic therapy poststroke.

Bupirone, a 5-HT_{1A} partial agonist of the azapirone class, is FDA-approved for anxiety and often used as an adjunct to SSRIs or serotonin-norepinephrine reuptake inhibitors, particularly in generalized anxiety disorder.⁷⁶ While it may represent a potential option in select cases, its efficacy and safety in PSA remain to be established. Data on first- and second-generation antipsychotics in anxiety disorders are also sparse, and their role in poststroke populations is unclear.⁷⁷

Gabapentinoids (gabapentin and pregabalin) share some pharmacological overlap with benzodiazepines and carry similar concerns regarding tolerance, dependence, addiction, and withdrawal.⁷⁸ They may offer benefit for anxiety symptoms, especially when coexisting with central poststroke pain, but current evidence is insufficient, and they are not considered first-line agents.⁷⁸

Noninvasive Brain Stimulation

Noninvasive brain stimulation uses magnetic pulses or electrical currents to modulate brain activity and is increasingly investigated as a treatment for PSD, particularly in patients who have not responded to standard pharmacological or psychological therapies or who cannot tolerate them due to side effects or contraindications. In the acute phase of stroke recovery, noninvasive brain stimulation is generally avoided due to cerebral vulnerability, but, in the chronic phases, it may be considered as an adjunct or alternative treatment option. Noninvasive brain stimulation is generally well-tolerated, with only mild and transient adverse effects such as headache or scalp discomfort, and its noninvasive, outpatient applicability enhances feasibility for clinical use. Among noninvasive brain stimulation modalities, repetitive transcranial magnetic stimulation and transcranial direct current stimulation are most widely studied.⁷⁹ A network meta-analysis suggests that repetitive transcranial magnetic stimulation significantly improves depressive symptoms both short- and long-term, and 20 minutes of 4-week dual-repetitive transcranial magnetic stimulation targeting the dorsolateral prefrontal cortex may yield superior outcomes compared with other protocols.⁸⁰ Despite promising findings, most existing trials are limited by small sample sizes, heterogeneity in protocols, and limited long-term follow-up.

Psychological and Behavioral Therapies

Several psychological and behavioral therapies have been adapted for use among people with stroke. Consistent with research on mental health after stroke, PSD has been the primary focus of psychological and behavioral therapies.⁵⁸ In general, the current research evidence would benefit from more methodologically rigorous studies using standardized intervention protocols, having a longer postintervention follow-up, and utilizing larger sample sizes that include people with communication and cognitive impairments, increasing generalizability to the stroke population.^{81,82}

Cognitive behavioral therapy (CBT) has been widely researched as a potential intervention for PSD.^{83,84} CBT works to improve psychological function through restructuring thoughts and changing maladaptive behaviors. Though early reviews were inconclusive,⁸⁵ a recent meta-analysis indicated significant benefits of CBT for managing PSA and PSD.⁸⁶ In addition, a smaller RCT found remotely administered CBT to be similarly effective as in-person,⁸⁷ which could make CBT more accessible for those who are geographically or situationally isolated.

Acceptance and commitment therapy is an empirically based psychological treatment aimed at enhancing psychological flexibility through acceptance, changing reactions to distressing thoughts and emotions, mindfulness, and self-awareness.⁸⁸ Group-based acceptance and

commitment therapy has shown promise with notable reductions in depressive symptoms^{89,90} but not for anxiety.⁹⁰ These preliminary results support further investigations of acceptance and commitment therapy to treat PSD.

The growth of mindfulness-based interventions in the treatment of anxiety and depression has contributed to interest in applying similar methods to stroke populations. Mindfulness is a state of mental being created by focusing on thoughts, feelings, and bodily sensations in the present moment without judgment.⁹¹ Early reviews of mindfulness-based interventions poststroke showed benefits to psychological function; however, recent reviews are more mixed.⁹² Interestingly, mindfulness-based interventions combined with cognitive therapy did demonstrate consistent benefits in reducing PSA and PSD,⁹² suggesting that mindfulness-based interventions may be useful as an adjunctive therapy.

Other promising psychological and behavioral treatments involve the application of positive psychology interventions. In contrast to traditional therapy models focused on fixing what is broken, positive psychology interventions build on existing strengths through promoting positive emotions and enhancing resilience. Early findings in stroke are promising, with decreases in depressive symptoms⁹³ and improvement in resiliency.⁹⁴

While many studies focus on evaluating either pharmacological or psychological interventions, few examine the combined benefits of both modalities.⁸¹ An RCT demonstrated superior effects of a brief psychosocial behavioral intervention combined with an antidepressant posttreatment and at 12-month follow-up compared with antidepressants alone.⁹⁵ These findings mirror treatment of depression across other populations, in which combined interventions show greater symptom improvement than medication or psychotherapy alone.⁹⁶

Other Nonpharmacological Approaches

Standard of care typically includes providing information; however, information provision alone is not particularly effective, and passive information provision may actually increase depressive and anxiety symptoms in stroke survivors.⁹⁷

Peer support approaches may be effective for stroke management and mental health support.⁹⁸ These types of interventions can be provided in hospital and community settings, and specific components vary considerably. A recent RCT with attention control demonstrated that a nurse-coled peer intervention held in community health centers decreased depression and anxiety symptoms, and also increased social participation, perceived social support, and reduced stigma, which are all important contributors to mental health.⁹⁹ Although results are promising, more rigorously designed, larger trials are needed to draw definitive conclusions.

Lifestyle modification approaches, such as diet and physical activity, can be valuable adjunctive approaches to pharmacological and behavioral interventions. According to a recent study, stroke survivors with 4 to 6 healthy lifestyle behaviors have an 80% lower risk of PSD, and each additional behavior is linked to a 45% risk reduction.¹⁰⁰ Numerous studies have shown the effectiveness of exercise in treating depression in the general population,^{101,102} and evidence supporting the use of exercise programming for PSD is equally compelling. Two meta-analyses of 24 RCTs (n=1757) found that exercise significantly alleviated depressive symptoms in patients with stroke compared with the control group.^{103,104} These meta-analyses also provided guidance on which types of programs may be most effective, favoring multicomponent training, which integrates various forms of exercise, including aerobic exercise, resistance exercise, and balance training, and significantly reduced symptoms over single-mode aerobic or resistance training, which demonstrated no reduction in symptoms.¹⁰⁴ In addition, the greatest benefit was demonstrated in interventions consisting of at least 3 sessions lasting <60 minutes per week or complying with American College of Sports Medicine recommendations (>30 minutes 5 days/week).¹⁰³ There is low evidence supporting mind-body interventions (ie, Tai Chi) as beneficial home-based exercise for PSD.¹⁰⁵

In addition to physical activity, a healthy diet and nutrition are considered an essential component of Life's Simple 7, which is inversely related to PSD risk.¹⁰⁶ Several recent studies based on National Health and Nutrition Examination Survey data demonstrate an association between vitamin B,¹⁰⁷ vitamin D,¹⁰⁸ and dietary antioxidant intake¹⁰⁹ and PSD. However, evidence based on rigorously designed trials to support the use of nutritional supplementation for PSA and PSD is lacking.

Music and art therapy have also attracted attention. These approaches may be used in conjunction with pharmacotherapies or psychotherapies to reduce anxiety and depressive symptoms by encouraging emotional expression and social interaction.¹¹⁰ They may be particularly useful for patients with cognitive or communication impairment. However, rigorous trials with large sample sizes are lacking to definitively support the benefit.

Caregiver and Dyadic Interventions

Many stroke survivors rely on a family caregiver, most often a spouse or partner, for support across various domains, reducing the cost of care and likelihood of rehospitalization.¹¹¹ Research indicates that anxiety and depression affect up to 40% of these caregivers¹¹² and are linked to increased burden, social isolation, and declines in health.¹¹³ Importantly, caregiver distress also negatively impacts the stroke survivor's recovery, potentially interfering with rehabilitation and increasing the risk

of rehospitalization.¹¹² Given the connection between stroke survivor and caregiver mental health, dyadic interventions have received growing interest over the past decade.¹¹⁴ A series of systematic reviews on individual caregiver interventions and dyadic interventions have repeatedly shown dyadic interventions to benefit stroke survivors for reducing depression and anxiety; however, findings on caregiver outcomes, including anxiety, depression, and caregiver strain, are more mixed.¹¹⁵

RESEARCH GAPS AND FUTURE DIRECTIONS

Despite growing recognition of PSD and PSA as important contributors to a range of outcomes, significant gaps remain in both research and clinical care.² Mental health issues poststroke continue to be underrecognized, and most available studies are limited by small sample sizes, short follow-up periods, and heterogeneous populations. While recent research has increasingly focused on PSD, the literature addressing anxiety, PTSD, pseudobulbar affect, alcohol and substance abuse, and psychosis remains especially scarce. Adding another layer of complexity, the absence of standardized diagnostic criteria, validated screening tools, consistent outcome measures, and clear definitions of remission or treatment response limits the comparability and generalizability of available studies, while, unfortunately, mental health metrics are rarely incorporated into stroke trials or only added as secondary outcomes and, therefore, are not appropriately powered. These limitations highlight the need for research across several areas, including a deeper understanding of PSD and PSA pathophysiology to guide targeted interventions; the development of neuroimaging, molecular, and neurophysiological biomarkers along with predictors of risk and treatment response; and clarification of the optimal timing for both pharmacological and nonpharmacological therapies across the acute, subacute, and chronic phases.¹¹⁶

PSD and PSA are unevenly distributed across population subgroups, reflecting disparities in burden, recognition, and access to care. Younger survivors face unique challenges as stroke disrupts work, finances, and family roles, yet rehabilitation programs remain oriented toward older adults and rarely include vocational or age-specific psychosocial support.¹¹⁷ Women consistently report higher rates of PSD and PSA, driven by older age at stroke onset, greater prestroke disability, and reduced access to secondary prevention combined with hormonal and inflammatory pathways, as well as less social support and lower socioeconomic resources, yet sex-sensitive approaches to screening and treatment remain underdeveloped.^{118,119} Racial and ethnic minorities experience disproportionate detection and treatment gaps, largely attributable to structural inequities in access to culturally

responsive mental health services and rehabilitation.^{120,121} Rural residents face travel barriers, social isolation, limited specialist access, and higher financial strain to get specialized services.¹²² Cognitive and communication impairment increases vulnerability to PSD and PSA but complicates detection, as standard screening tools often fail in this group, and research frequently excludes them, biasing prevalence estimates downward.^{123,124} Addressing these disparities will require inclusive study designs, culturally and linguistically adapted assessment tools, and policies that expand access through digital interventions (including telehealth, virtual reality, mobile apps, and Internet-based platforms), community-based resources, and a diverse and informed mental health workforce.

Finally, it is important to emphasize the value of interdisciplinary collaboration, particularly the integration of mental health professionals into stroke care. Neurologists, psychiatrists, and other stroke providers should play a leading role in recognizing and managing PSD and PSA while ensuring coordination with psychiatry, psychology, and rehabilitation teams. Establishing clear referral pathways and embedding mental health support within inpatient and outpatient programs can facilitate more timely, effective, and patient-centered management.

ARTICLE INFORMATION

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Sources of Funding

None.

Disclosures

Dr Terrill reports funding from the National Institutes of Health Eunice Kennedy Shriver National Institute of Child Health and Human Development, and compensation from the American Heart Association for speaker honorarium. Dr Mackenzie reports funding from the National Institutes of Health. The other authors report no conflicts.

Supplemental Material

Tables S1–S2

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